products_05-kheti-mitra_ORDERING

5 Kheti-Mitra irrigation: ordering guide (what to buy, where, in what order)

> Companion to MANUFACTURING_MANUAL.md. Main caveat for this project: field trial required. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: manuals/ORDERING_GUIDE.md.

1. Order of work

1. Read the manual and the caveat above.
2. Order the circuit boards and SMD assembly (section 2).
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

2. Order the circuit boards (JLCPCB, PCB + assembly)

Files (all in this repository):

- Gerbers: `utility10/pcb/release/kheti/1_fabrication/kheti_mitra_gerbers.zip`
- Assembly BOM: `utility10/pcb/release/kheti/2_assembly/kheti_mitra_JLC_BOM.csv`
- Pick-and-place (CPL): `utility10/pcb/release/kheti/2_assembly/kheti_mitra_JLC_CPL.csv`
- Full BOM with LCSC numbers and prices: `costing/kheti_priced_BOM.csv`

Steps:

1. Create an account at **jlcpcb.com** and click *Order now*. Upload the Gerber ZIP. The viewer must show the board outline, both copper layers, silkscreen and drill holes.
2. Settings: **2 layers, 1.6 mm, FR-4**, lead-free HASL (use **ENIG** if the board has an LGA/QFN fine-pitch part, e.g. ADXL345, ESP32 module pad, BQ25570), green mask, white silk, quantity 5 (the minimum), *Remove order number: specify a location* if you want to avoid the order-number mark. Heavier copper (2 oz) only where the board notes say so.
3. Switch on **PCB Assembly**: *Economic*, assembly side *Top*, and set tooling holes *Added by JLCPCB* unless the board already has them. Choose the number of boards to assemble (2 or 5).
4. Upload the BOM and CPL. In the parts step, every line must match (LCSC numbers are in the BOM). Where a part is shown as *out of stock* or *no match*, use the alternative suggested by JLCPCB only after checking footprint and value against the list below; do not accept a different footprint.
5. In the **component placement preview**, check pin 1 / polarity of every IC, diode, LED, electrolytic capacitor, USB-C and connector against `*_assembly_top.pdf` in the same folder. Wrong rotation is the most common assembly error: ask JLCPCB support to rotate before approving.
6. Pay, then approve the assembly engineering review messages (they usually ask about a part or rotation). Delivery by DHL/FedEx/UPS takes about 1 to 2 weeks to India.

Parts **not** assembled by JLCPCB (you buy and solder them yourself):

Refs	Value	Part	LCSC
J2	ISP	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964
J1	12 V DC in	CUI/Same Sky PJ-063AH	C22434582
J5	Valve 12V	Cixi Kefa KF128-5.08-2P-AA	C474952
J6	Soil sensor 1	JST B3B-PH-K-S(LF)(SN)	C131339
J7	Soil sensor 2	JST B3B-PH-K-S(LF)(SN)	C131339
J8	Rain sensor	JST B3B-PH-K-S(LF)(SN)	C131339
TP1, TP2, TP3, TP4	TP		

Order these with the same LCSC order (add them to the cart so one shipment arrives) or buy locally in India (section 5).

Parts cost and stock (live LCSC prices on the run date)

Total parts per board: **US6.56*for1board* * *, * * *US*5.83 each for 10 boards**, **US$4.80 each for 100 boards** (about INR 577 / 513 / 422 at 88 INR/USD). The per-line break-up is in `costing/kheti_priced_BOM.csv`. This is parts only: PCB fabrication, assembly fees, shipping, customs and GST come on top (see `manuals/ORDERING_GUIDE.md`, section 6).

Stock warnings on the run date:

- TP (no lcsc price (hand-fit or not priced), stock)

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
U1	1	ATmega328P-AU	ATMEGA328P-AU	C14877	2.3308	2.331
C1,C2,C3,C8,C9,C10,C11	7	100n	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	0.0075	0.052
R7,R23	2	10k	0603 10k 1%	C25804	0.0024	0.005
J2	1	ISP	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964	0.0766	0.077
D1	1	Green	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	0.0105	0.011
R5,R20,R21,R22	4	1k	0603 1k 1%	C21190	0.0017	0.007
J1	1	12 V DC in	CUI/Same Sky PJ-063AH	C22434582	2.7602	2.760
F1	1	PTC 1A 16V	Littelfuse 1812L110/16DR (1.1 A hold, 16 V)	C142746	0.1050	0.105
D3	1	SMBJ15A	Littelfuse SMBJ15A	C83846	0.1300	0.130
Q3	1	AO3401A	AO3401A	C15127	0.0969	0.097
R17,R18	2	100k	0603 100k 1%	C25803	0.0030	0.006
U3	1	AMS1117-5.0	AMS AMS1117-5.0 SOT-223	C6187	0.2602	0.260
C6,C7	2	10u	0805 10uF X5R	C15850	0.0441	0.088
Q4	1	AO3400A	AOS AO3400A	C20917	0.0899	0.090
R16	1	100	UNI-ROYAL 0603WAF1000T5E	C22775	0.0024	0.002
D7	1	SS14	Starsea SS14 SMA	C84343	0.0252	0.025

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
J5	1	Valve 12V	Cixi Kefa KF128-5.08-2P-AA	C474952	0.1948	0.195
J6	1	Soil sensor 1	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J7	1	Soil sensor 2	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J8	1	Rain sensor	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
SW2	1	Hold	Omron B3U-1000P	C231329	0.1894	0.189
TP1,TP2,TP3,TP4	4	TP				

2b. Enclosure: printed parts and fasteners (generated from the board)

- Print `4_enclosure/` base and lid (outer about 101 x 75 x 23 mm) in PETG or ABS, 3 perimeters, 25 % infill; the lid closes with **4 x M2.5 x 10 mm self-tapping screws** into the corner columns; the board sits on its standoffs with 4 x M2.5 x 6 mm self-tapping screws.
- Print `*_enclosure_caps.stl` (light pipe, plunger): plunger for SW2; light pipe for D1. Light pipes in clear PETG or resin; fit plungers and pipes from inside before closing the lid.
- indoor, dry use; print in PETG or ABS, 3 perimeters, 25 % infill
- Print one set first and test-fit the board, the holder and every cap; the dimensions are first-pass.

3. Bought-in and mechanical items

2. Bill of materials

- Board parts (37 parts incl. barrel jack, polyfuse, TVS, AMS1117-5.0, valve MOSFET AO3400A, Schottky flyback diode SS14): see the table in `utility10/pcb/release/kheti/README.md` and the machine-readable `2_assembly/kheti_mitra_BOM_full.csv`.
- Bought-in: 12 V DC adapter (BIS-registered, 1 A or more), 12 V normally-closed solenoid valve (1/2 in), 2 capacitive soil-moisture sensors (3-wire, 3.3-5 V), rain sensor module (digital output), tubing/fittings, IP65 box if used outdoors, cable glands.
- Designed production BOM is Rs 620 (earlier guess Rs 520); recommended price Rs 2,299 (see product README economics).

4. Confidence (read before you order)

- **Order confidence: Medium-High**: limited stock on: J1 12 V DC in (stock 321).
- **Works-as-intended confidence: Medium**: Irrigation controller with 12 V switching and soil sensing; 22 host checks pass; field conditions (water, sun, surge) are the risk.
- **First thing to do:** Run with a lamp as the load for 48 hours before any valve or pump.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts	**LCSC** (lcsc.com) with the same cart as JLCPCB
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.